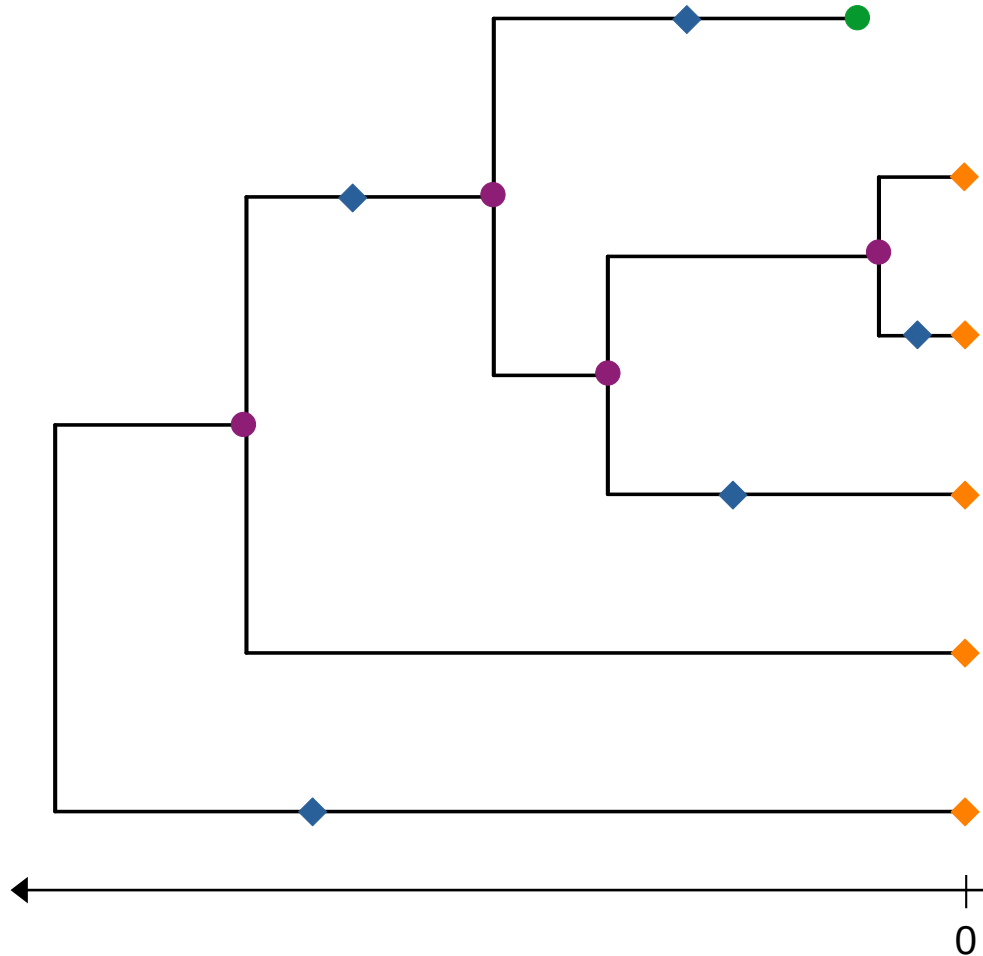


Integrating fossil evidence into phylogenetic dating Extensions and limits

Joëlle Barido-Sottani

The fossilized birth-death (FBD) model



Parameters:

- λ — birth rate
- μ — death rate
- ◆ ψ — fossil sampling rate
- ◆ ρ — extant species sampling probability

Fossils in Bayesian inference

$$P(\text{[Posterior components]} | \text{[Likelihood components]}) =$$

Posterior

Likelihood

Probability of
the tree model

Priors

$$P(\text{ACAC... TCAC... ACAG...} | \text{[Substitution model] [Clock model] [Time tree]})$$

$$P(\text{[Fossil] [Time tree]} | \text{[Tree model]})$$

$$P(\text{[Substitution model] [Clock model] [Tree model]})$$

$$P(\text{[Fossil]} | \text{ACAC... TCAC... ACAG...})$$



Fossil ages

ACAC...
TCAC...
ACAG...

Molecular alignment



Substitution model



Clock model



Time tree

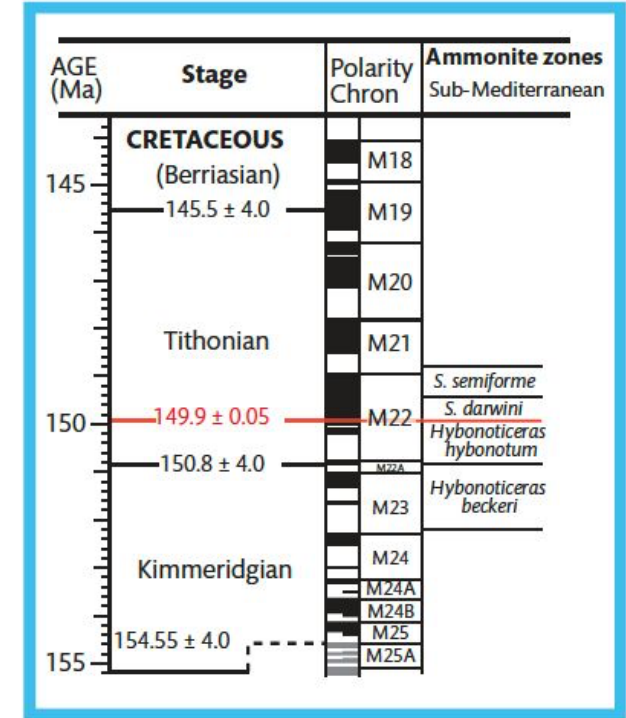
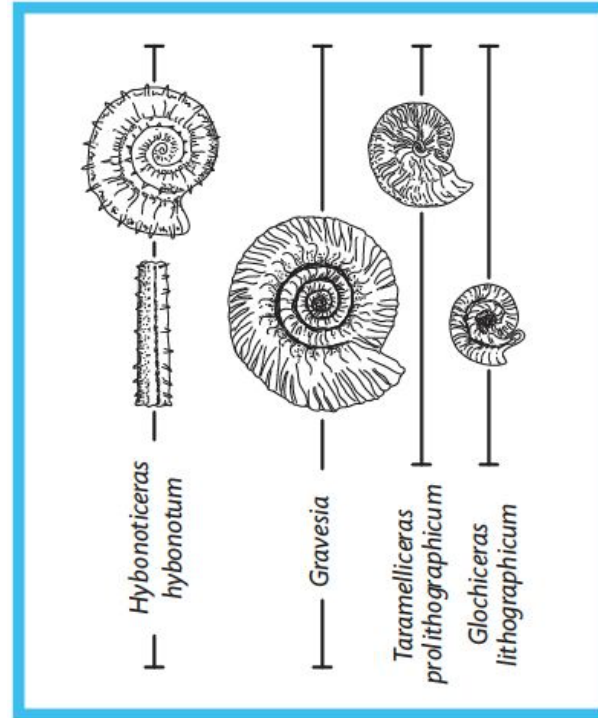
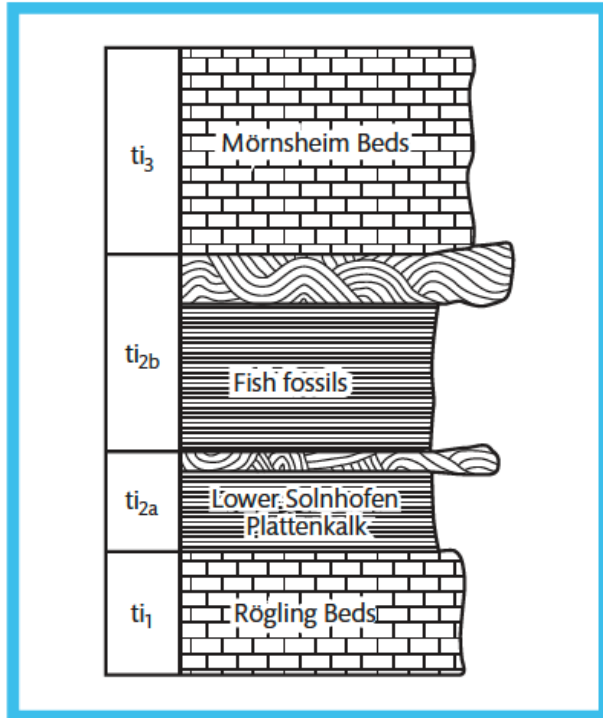


Tree model

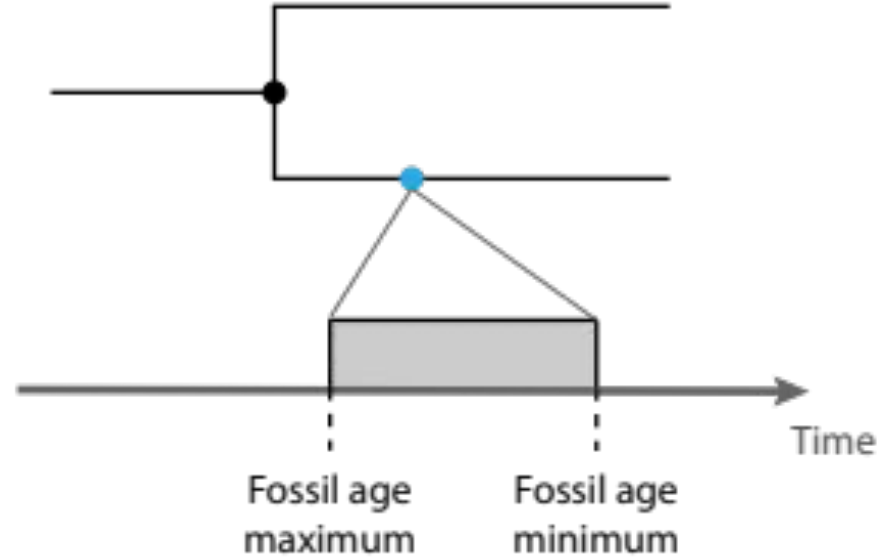
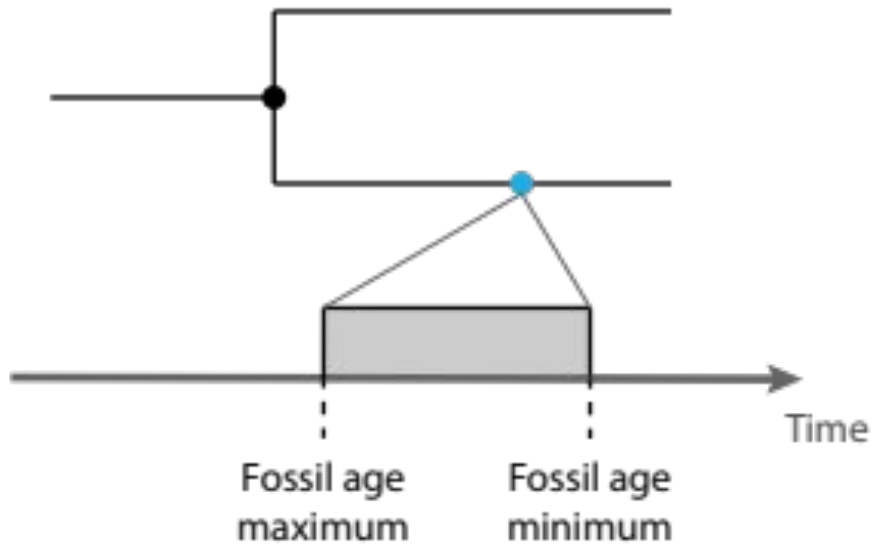
Outline

- Integrating new sources of information
 - Fossil age uncertainty
 - Occurrence data
 - Diversified sampling for large clades
- Correcting and extending model representations
 - Stratigraphic ranges
 - Variations through time
 - Rate heterogeneity

Fossil age uncertainty



Integrating the uncertainty



Fossil age uncertainty can be
sampled as part of the MCMC

Age uncertainty in Bayesian inference

$$P(\text{Genetic data} \mid \text{Tree model} \mid \text{Fossil age ranges}) =$$

Posterior

Likelihood

Probability of
the tree model

Priors

$$P(\text{ACAC... TCAC... ACAG...} \mid \text{Genetic data}) \times P(\text{Tree model} \mid \text{Fossil age ranges}) \times P(\text{Genetic data} \mid \text{Tree model})$$

$$P(\text{Fossil age ranges} \mid \text{ACAC... TCAC... ACAG...})$$



Fossil age ranges

Integrating the uncertainty

Ignoring stratigraphic age uncertainty
leads to erroneous estimates of species
divergence times under the fossilized
birth – death process

Joëlle Barido-Sottani^{1,2,3}, Gabriel Aguirre-Fernández⁴, Melanie Hopkins⁵,
Tanja Stadler^{1,2} and Rachel Warnock^{1,2,4}

Fossil age uncertainty **should** be
sampled as part of the MCMC

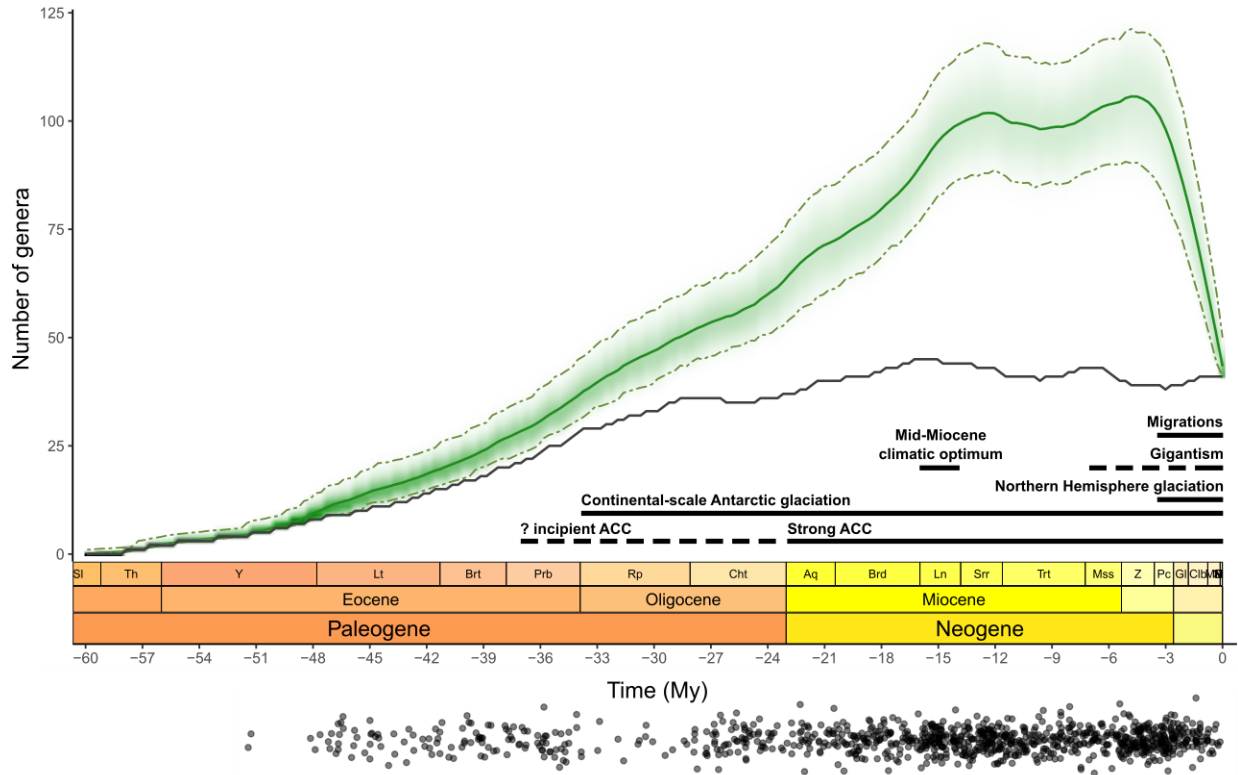
Integrating occurrences

Occurrence = sample without associated sequence information

=> Database of fossil specimens

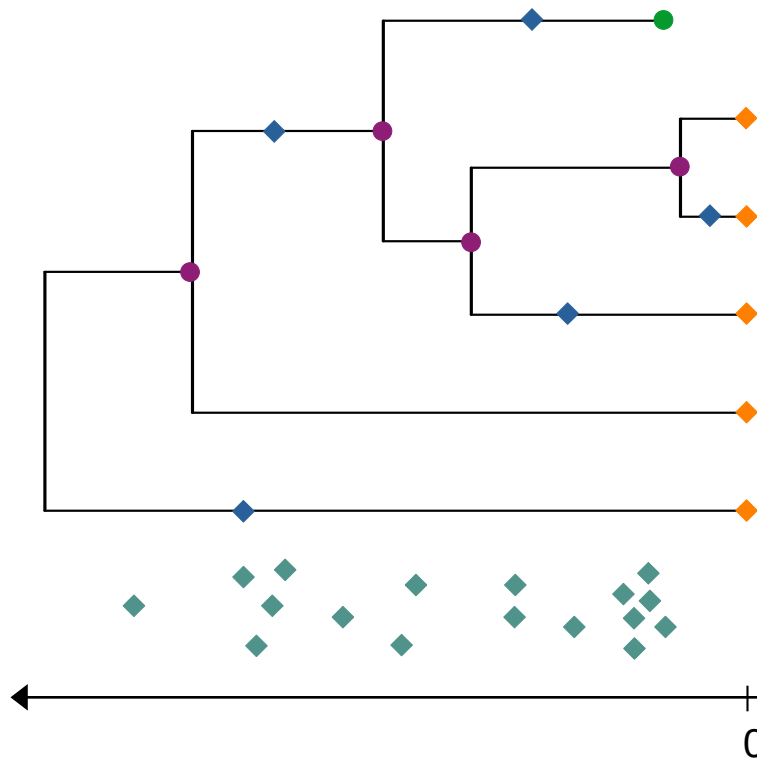


The Paleobiology Database
revealing the history of life



Occurrence birth-death process (OBDP)

Using occurrences to infer the total number of lineages through time and inform the population parameter estimates



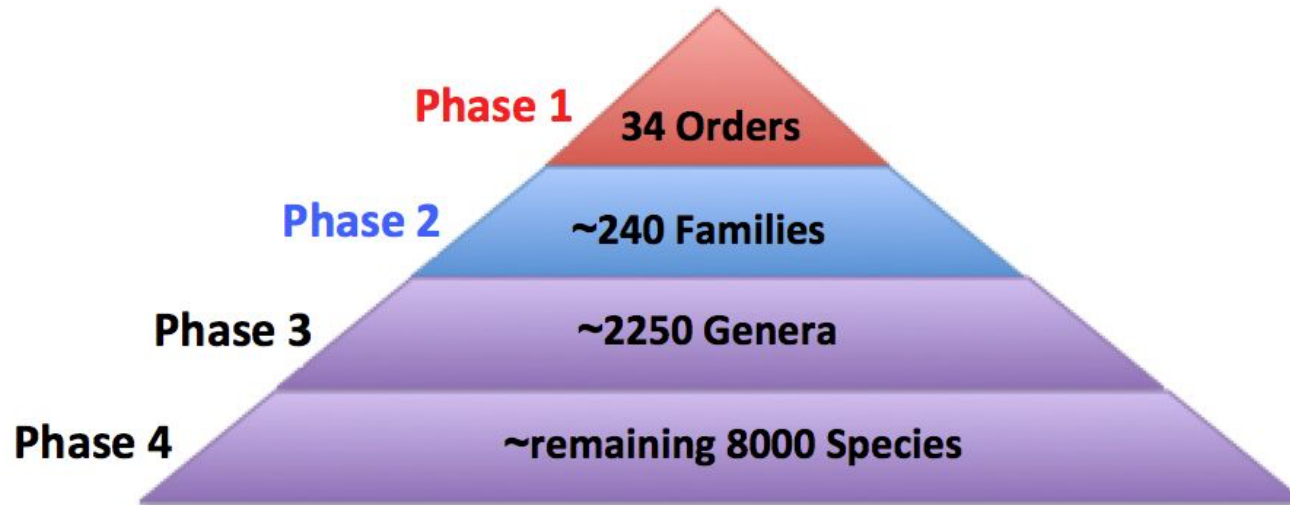
- λ — birth rate
- μ — death rate
- ◆ ψ — fossil sampling rate
- ◆ ρ — extant species sampling probability
- ◆ ω — occurrence sampling rate

Gupta et al. (2020), Manceau et al. (2021),
Andréoletti et al. (2022), Zarebski et al. (2023)

The problem of size

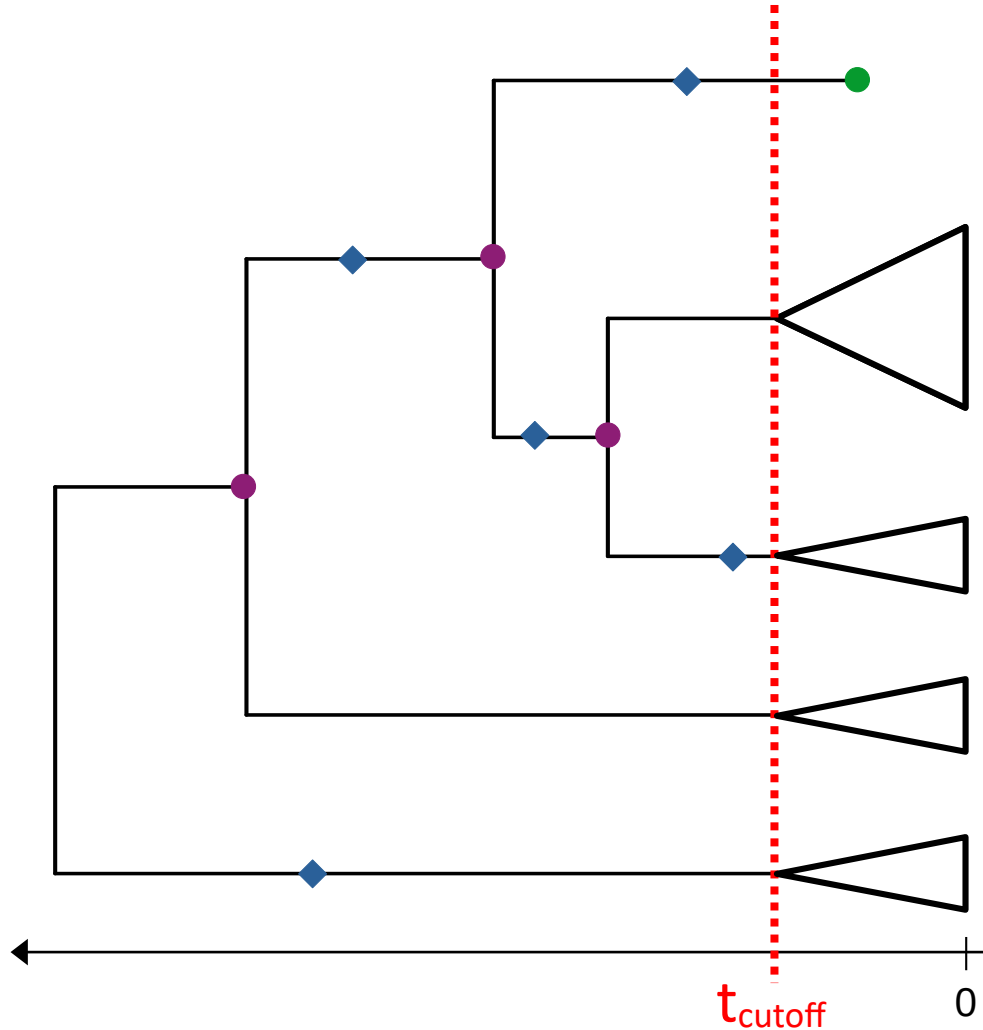
Bayesian FBD inference : up to ≈ 500 -800 samples

VS



Source : Bird 10,000 Genomes (B10K) Project

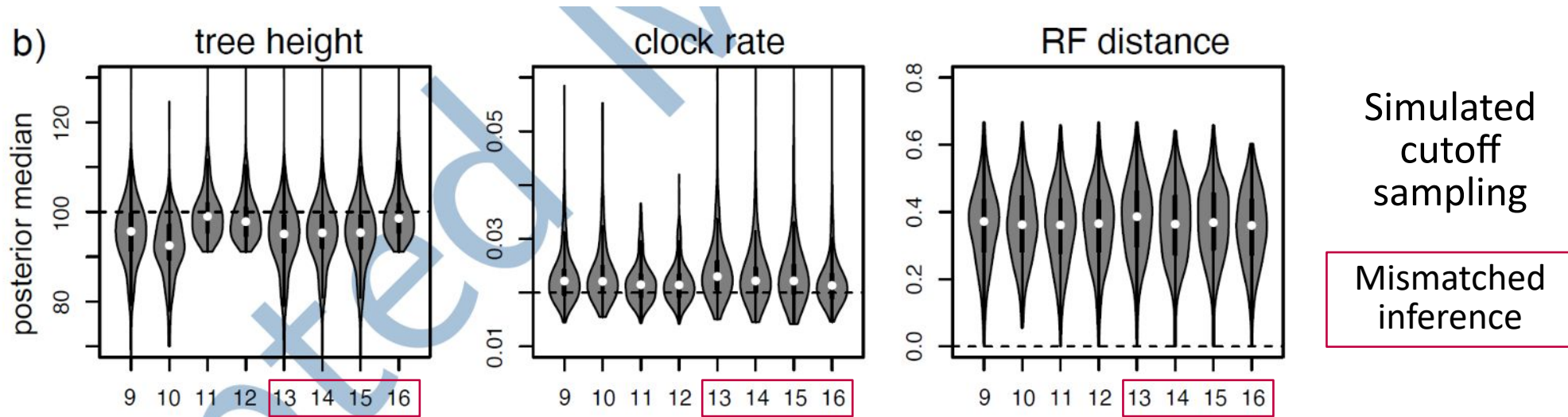
FBD model with diversified sampling



Parameters:

- λ — birth rate
- μ — death rate
- ◆ ψ — fossil sampling rate
- ◁ N — total number of extant species

Does the sampling process matter?



Zhang et al. (2023)

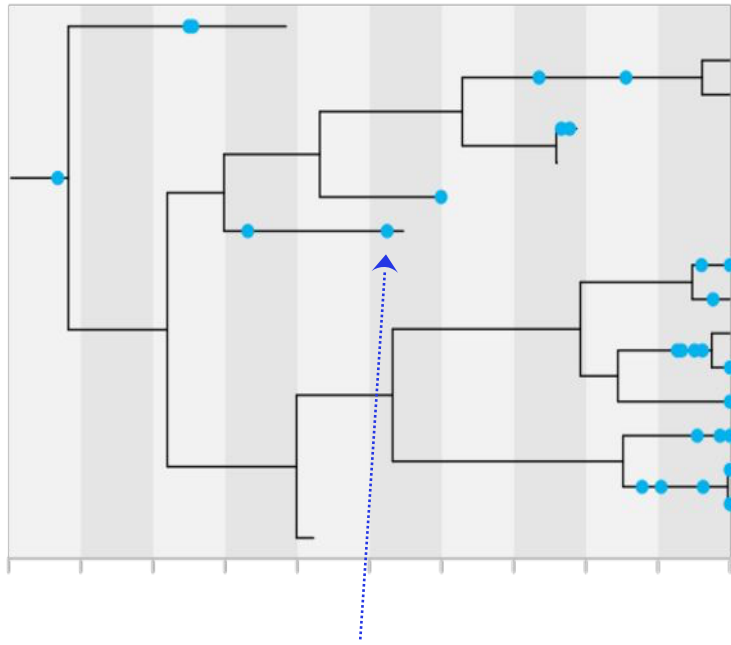
Caveats:

- Diversification rates are biased by sampling mismatch
- Simulated trees tend to be more balanced than real trees

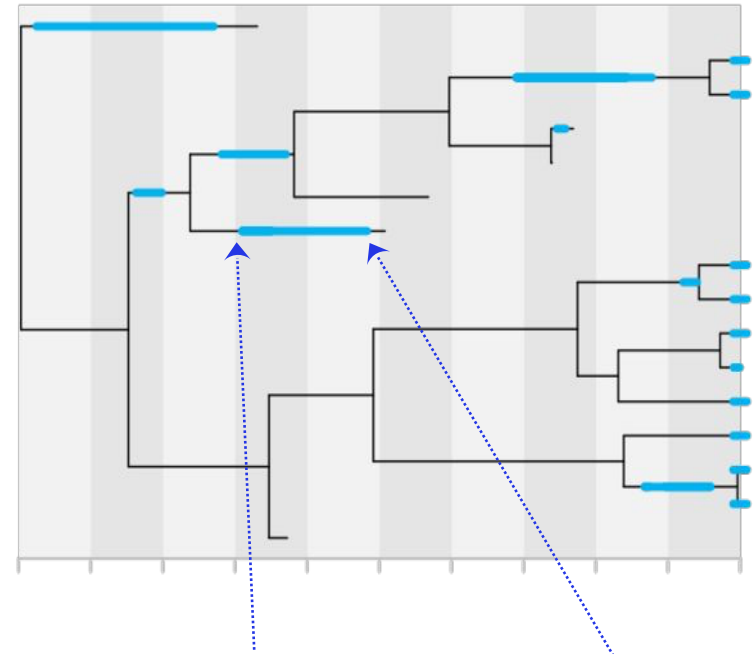
Outline

- Integrating new sources of information
 - Fossil age uncertainty
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 - Variations through time
 - Rate heterogeneity

Specimen-level data vs range data



Specimen (one
occurrence)

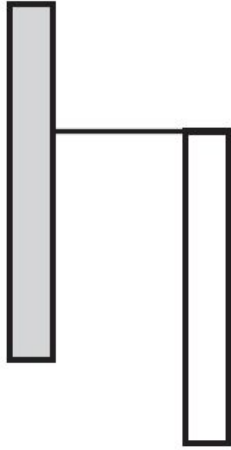


First occurrence

Last occurrence

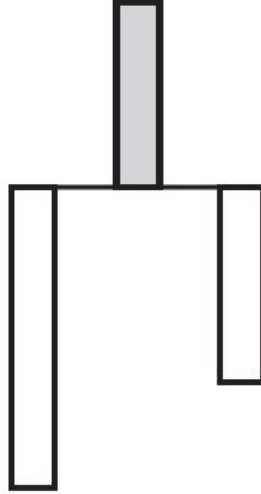
Fossil species $\neq$ Tree lineages

(i) asymmetric speciation (ii) symmetric speciation (iii) anagenetic speciation



1 birth event

1 speciation event



1 birth event

2 speciation events
1 extinction event



0 birth event

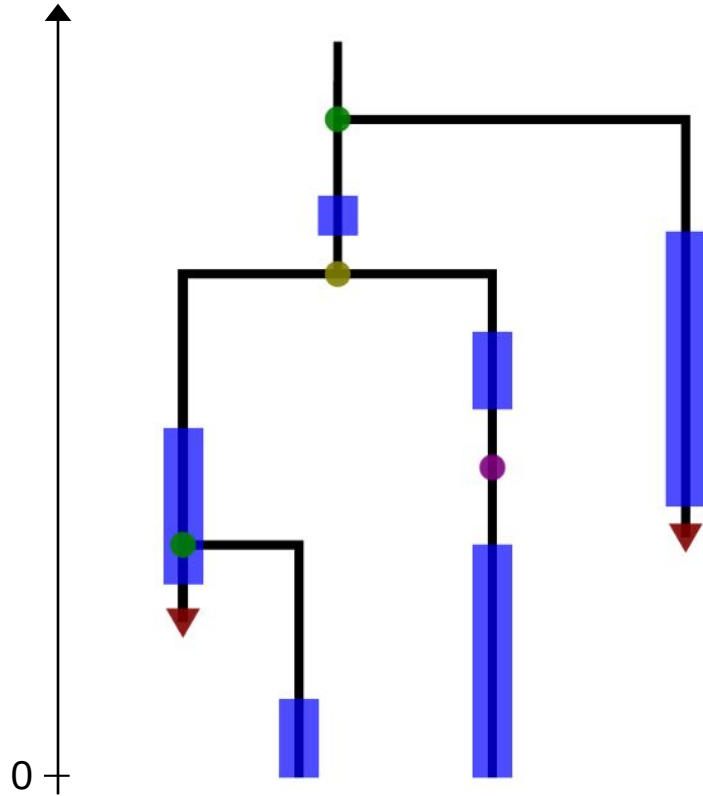
1 speciation event
1 extinction event

The FBD for stratigraphic ranges



FBD-range model

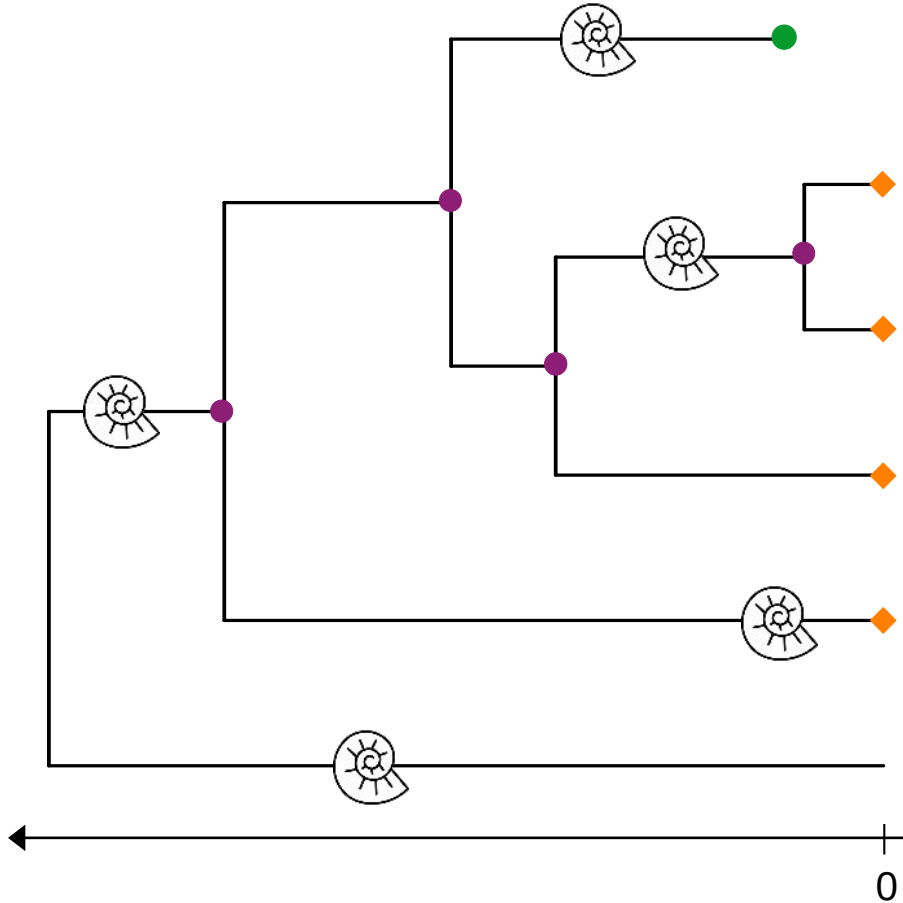
- Budding speciation
- Bifurcating speciation
- Anagenetic speciation
- ▼ Extinction



Remaining challenges

- Full implementation
- Identifiability of the different processes
- Combination with morphological data
-

The (homogeneous) FBD process

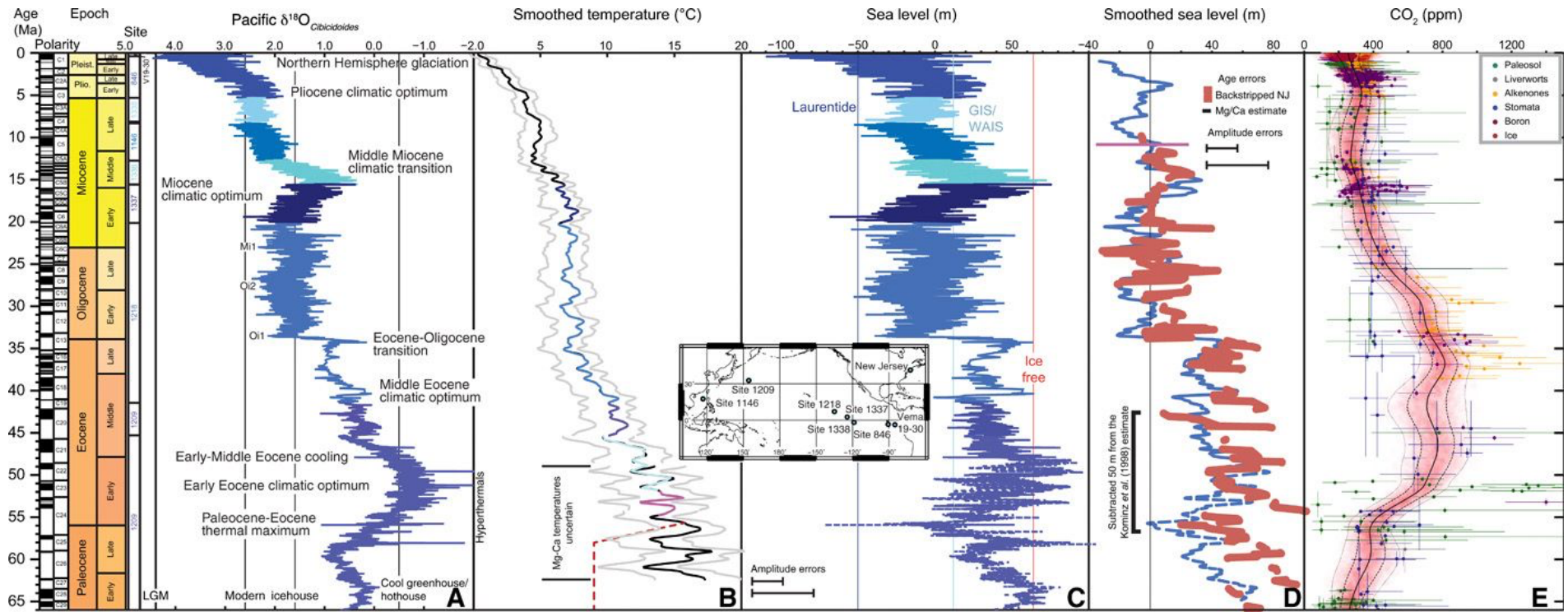


Parameters:

- λ — (constant) birth rate
- μ — (constant) death rate
- 🌀 ψ — (constant) fossil sampling rate
- ◆ ρ — extant species sampling probability

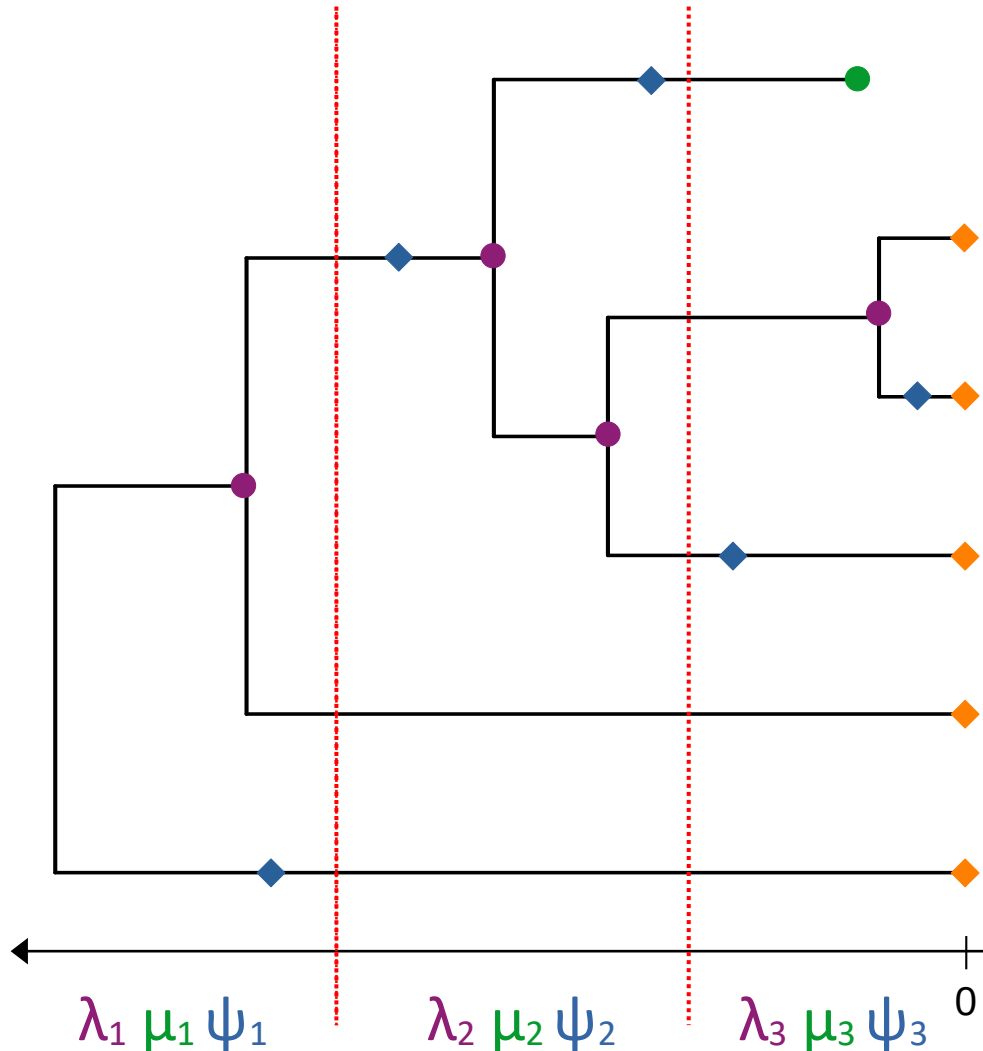
Variations in time

Many factors change through time: continents, climatic conditions, sea levels, etc.



Miller *et al.* (2020)

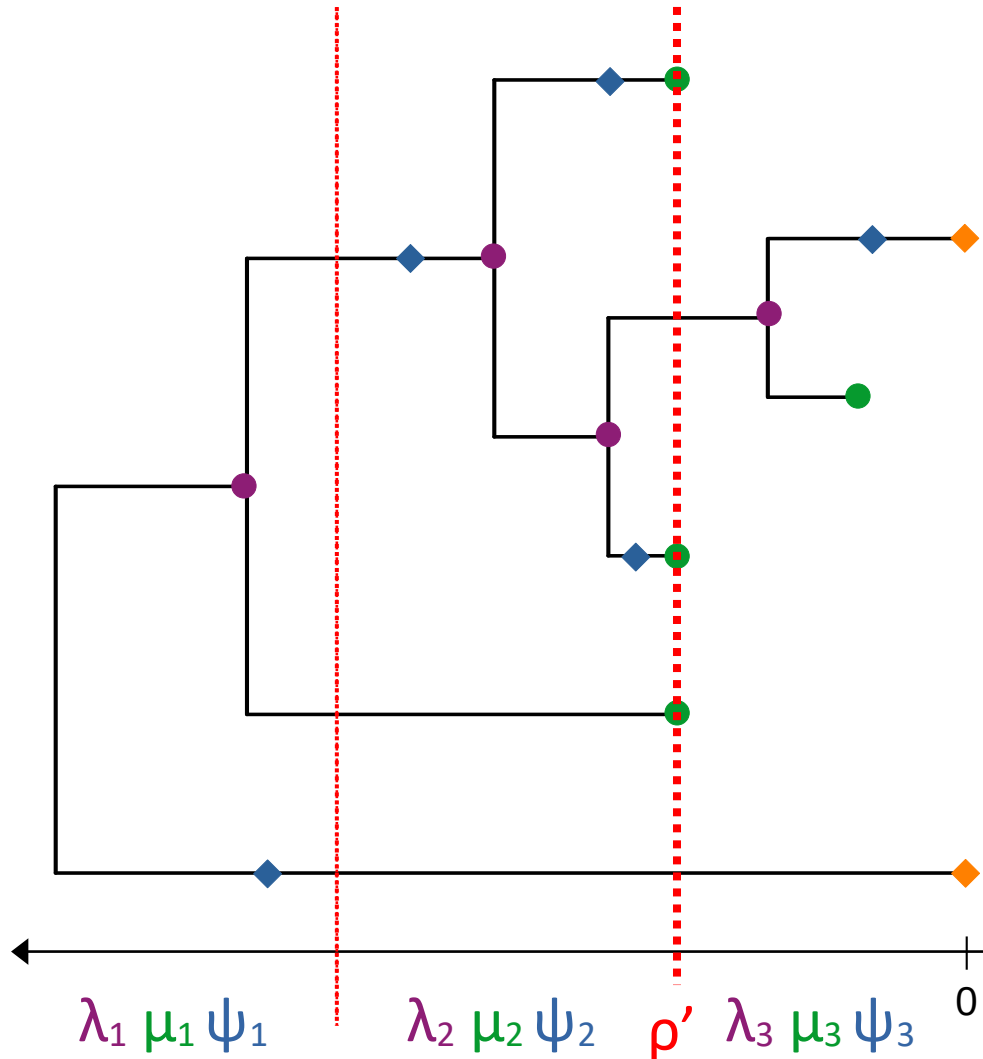
Skyline FBD model



- Allows for punctual shifts in rates through time
=> piecewise-constant rates
- Interval boundaries can be
 - fixed to specific ages
 - fixed to specific events (e.g. tree origin)
 - inferred by the method

Stadler (2011)

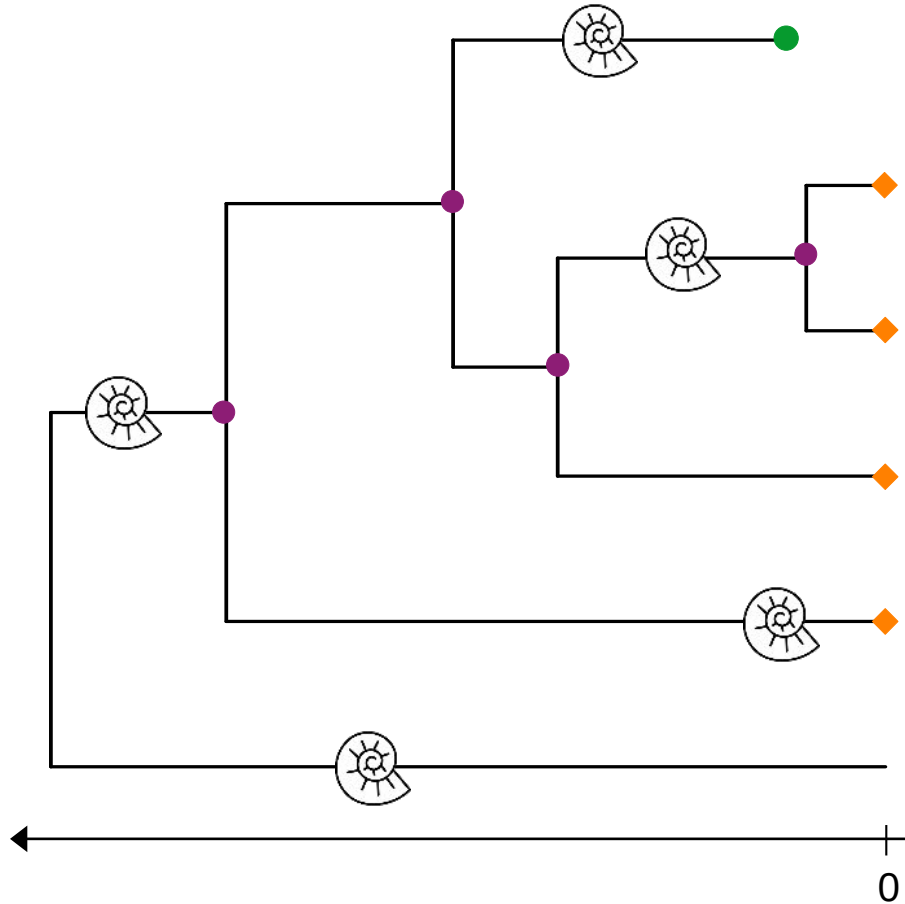
Mass extinction events



Parameters:

- λ — birth rate
- μ — death rate
- ◆ ψ — fossil sampling rate
- ◆ ρ — extant species sampling probability
- ρ' — mass extinction probability

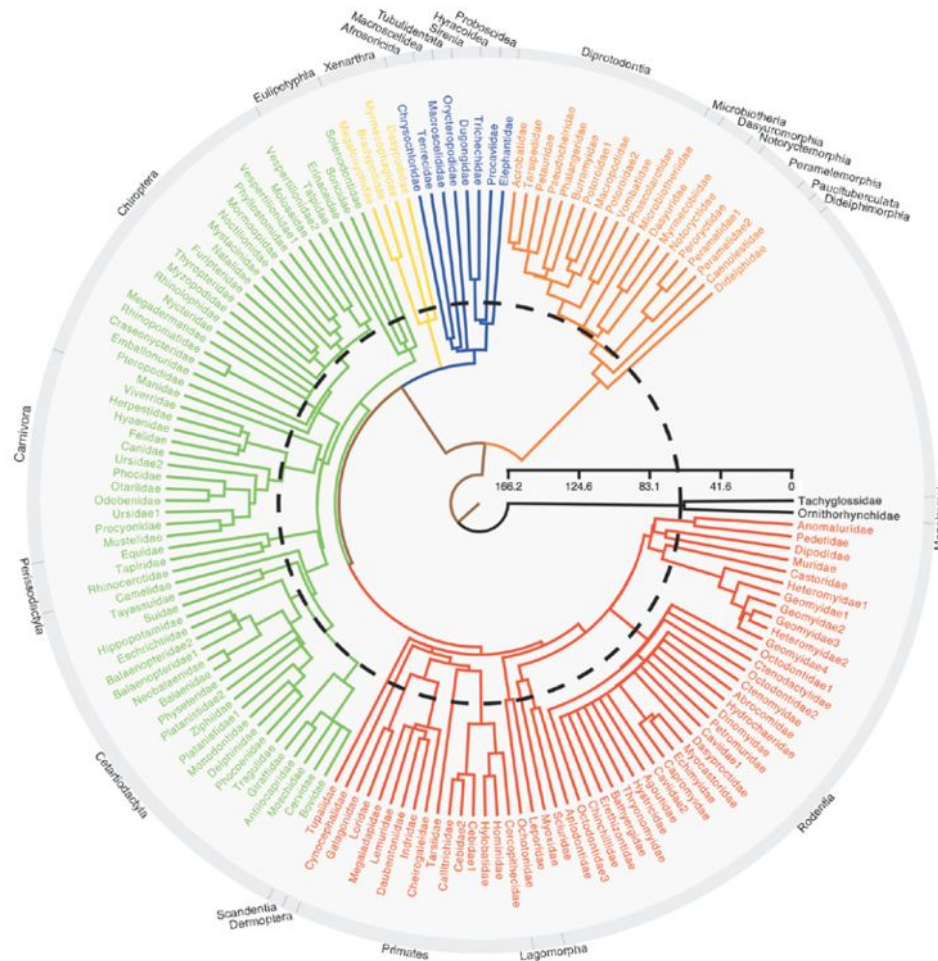
The (homogeneous) FBD process



Parameters:

- λ — (constant) birth rate
- μ — (constant) death rate
- 🐚 ψ — (constant) fossil sampling rate
- ◆ ρ — extant species sampling probability

Diversification – an homogeneous process ?

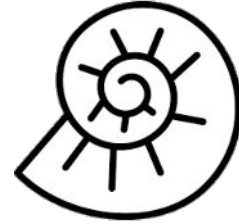


Traits proposed as driving rate variations :

- body size
- environment
- mating system

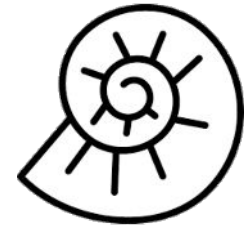
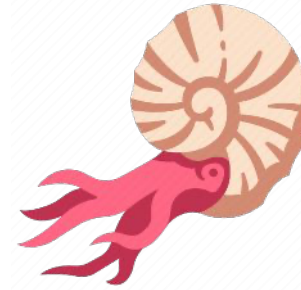
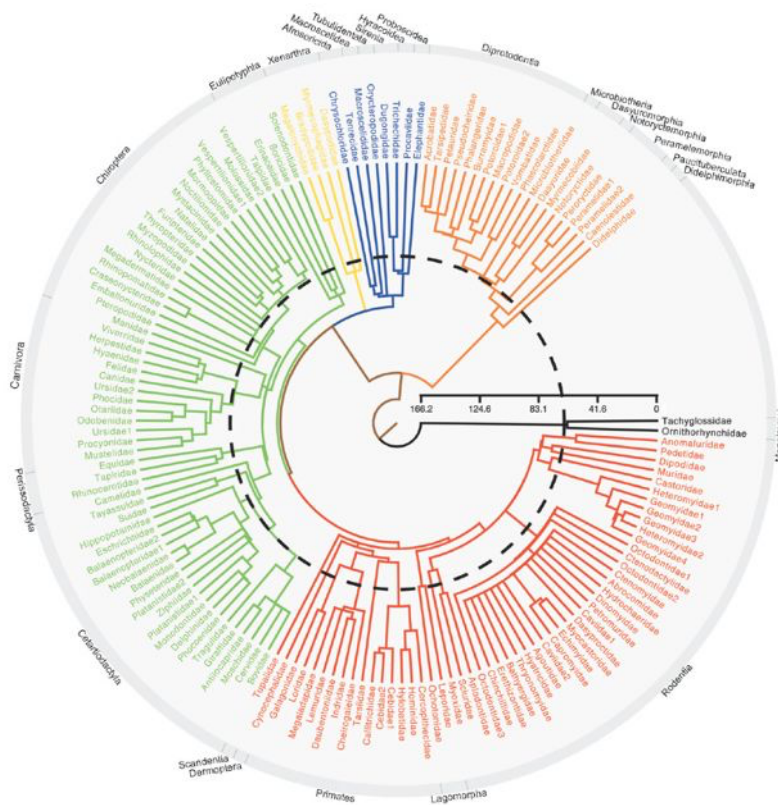
And many others.....

Fossilization – an homogeneous process ?

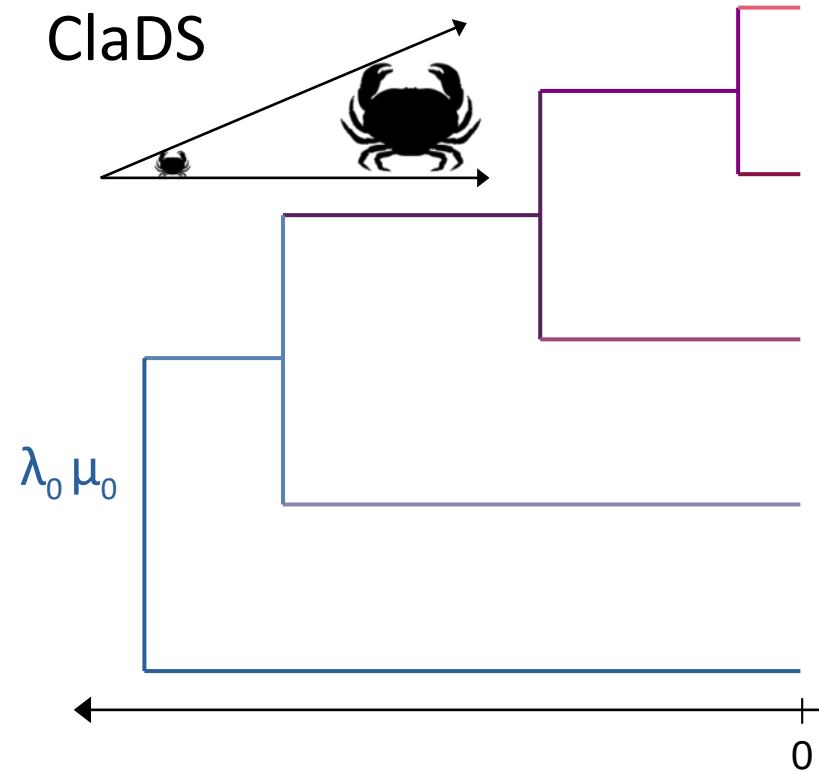
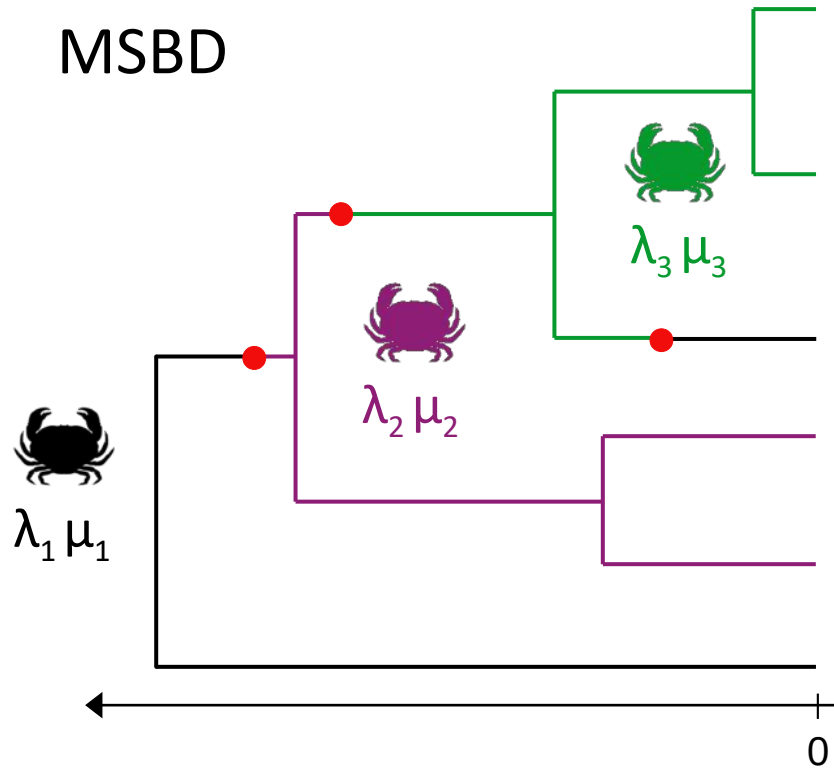


- Taphonomic biases: soft tissue degradation, environmental and time differences in sedimentation
- Sampling biases: geographical differences, identification issues

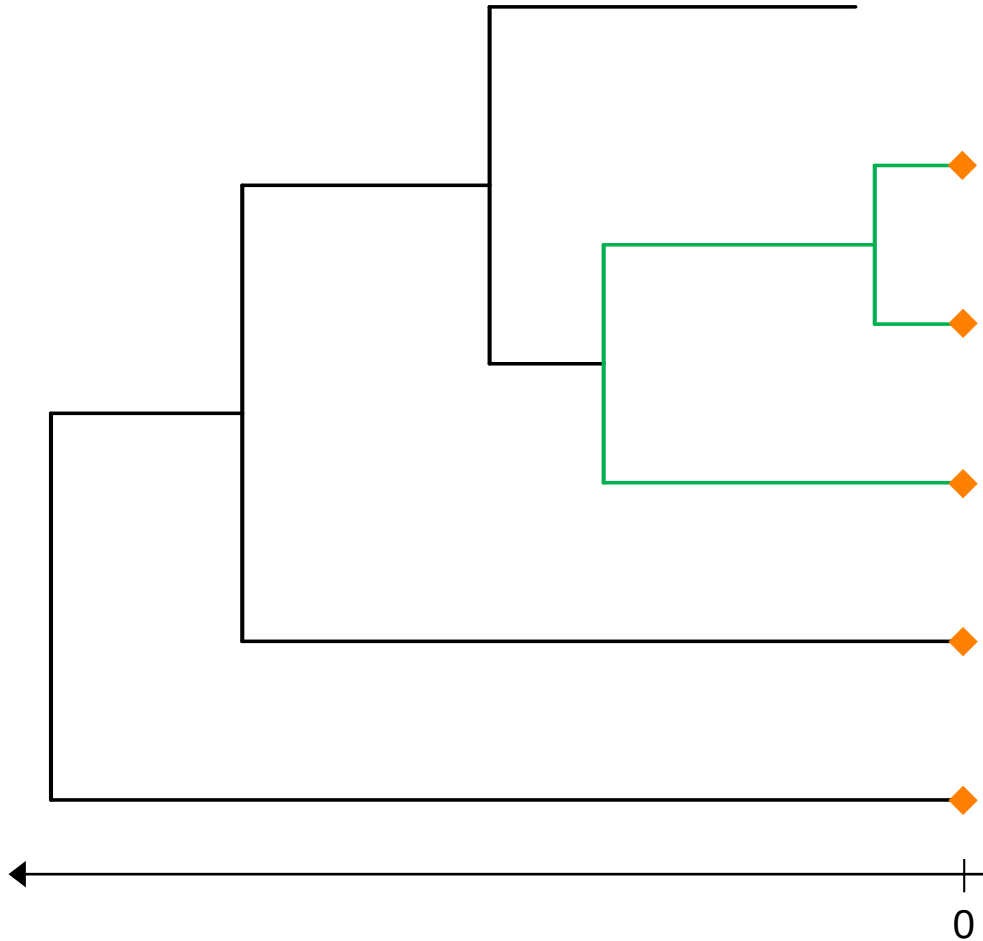
How can we integrate these variations in our inferences ?



Heterogeneous BD models



Multi-type birth-death (MTBD) process



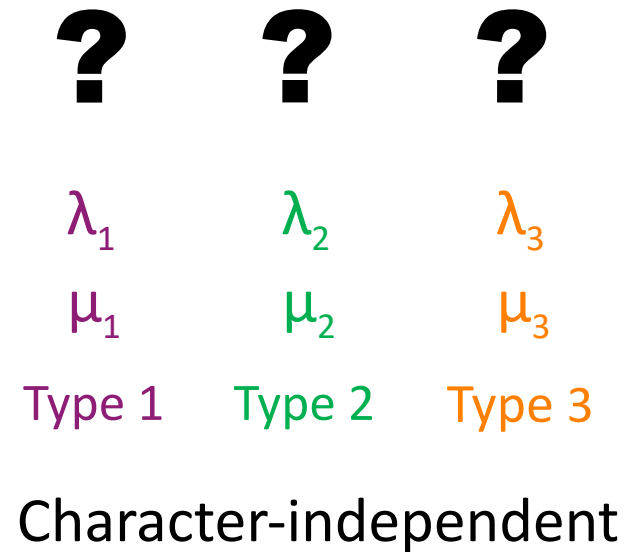
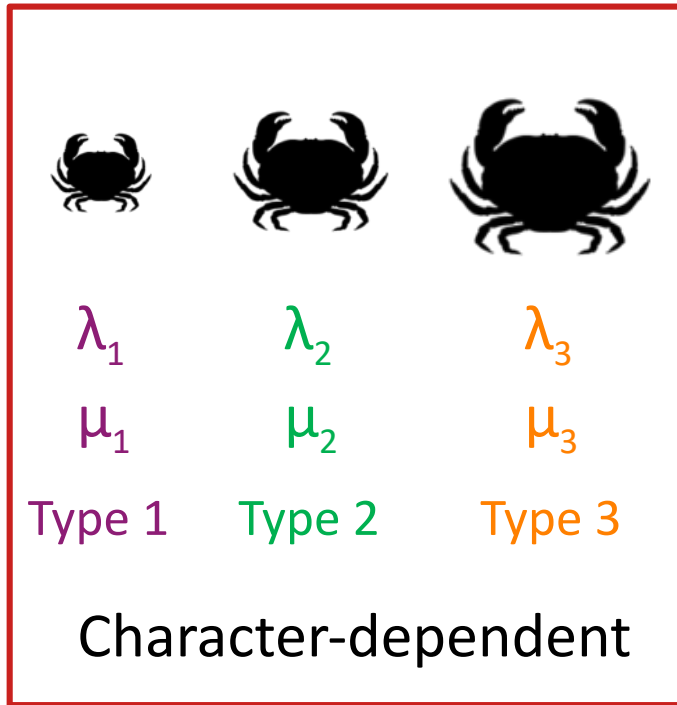
2 types, type 1 & **type 2**

λ_1 & **λ_2** — birth rates

μ_1 & **μ_2** — death rates

ρ — extant species
sampling probability

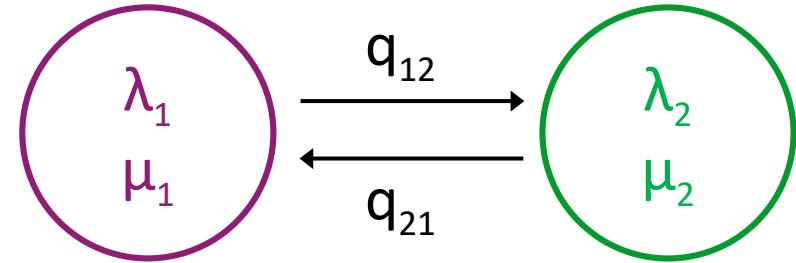
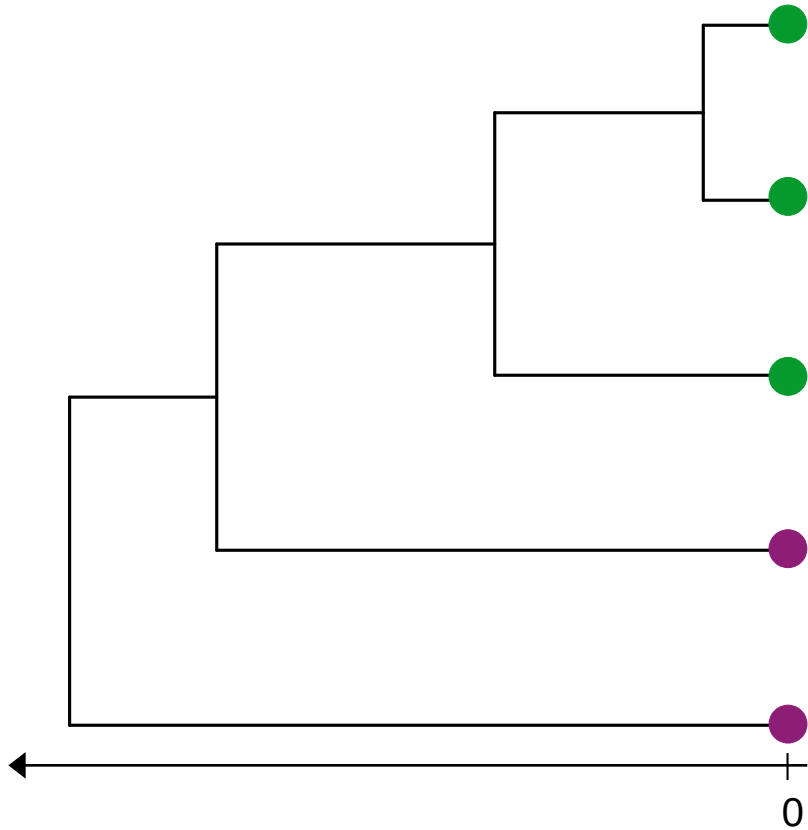
Character-dependent or independent ?



In a character-dependent model :

- The number of types is known
- The type at the tips is known

The BiSSE/MuSSE/BDMM model



Parameters of the model:

λ_i – birth rates

μ_i – death rates

q_{ij} – transition rates

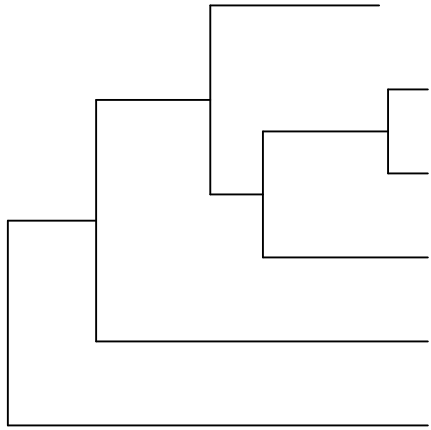
ρ/p – sampling probability

Maddison *et al.* **Sys. Bio.** 2007

Fitzjohn *et al.* **Sys. Bio.** 2009

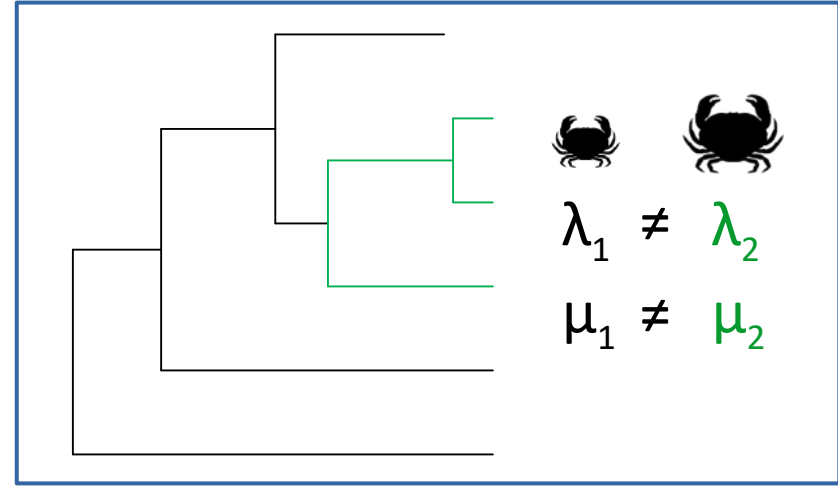
Kühnert *et al.* **MBE** 2016

Model selection issues



$$\lambda_1 = \lambda_2$$

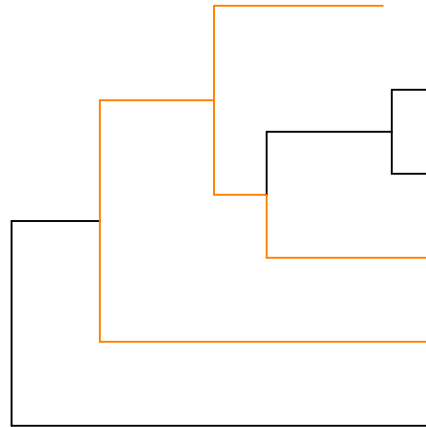
$$\mu_1 = \mu_2$$



$$\lambda_1 \neq \lambda_2$$

$$\mu_1 \neq \mu_2$$

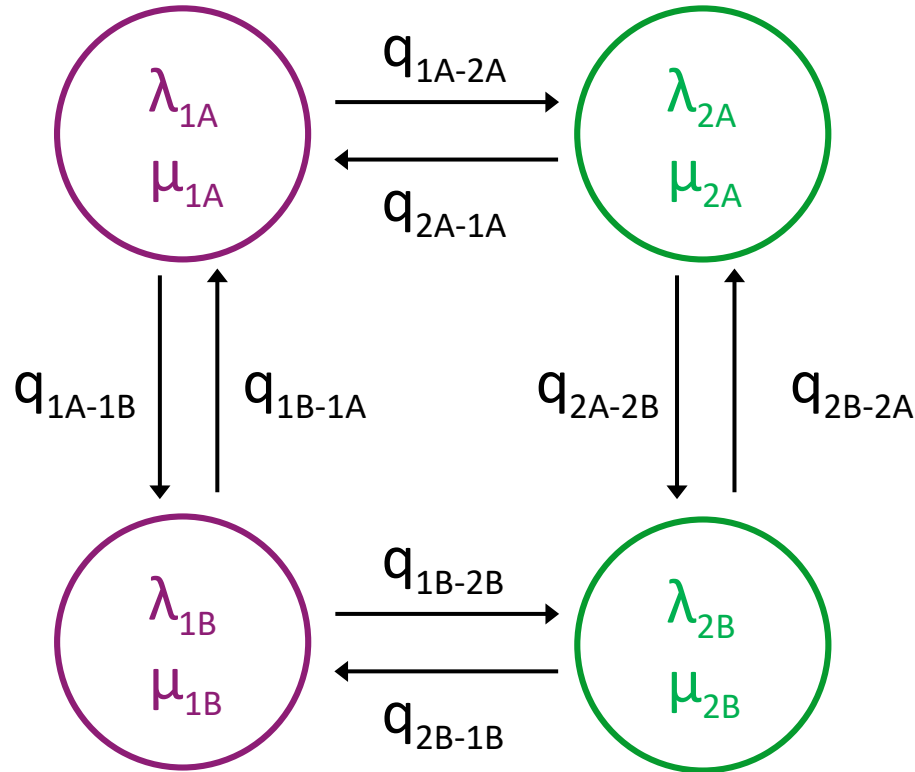
Rabosky & Goldberg 2015, Sys. Bio.



$$\lambda_1 \neq \lambda_2$$

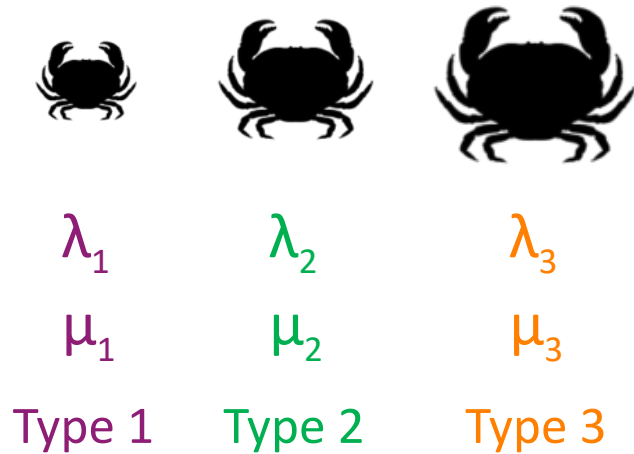
$$\mu_1 \neq \mu_2$$

The HiSSE model

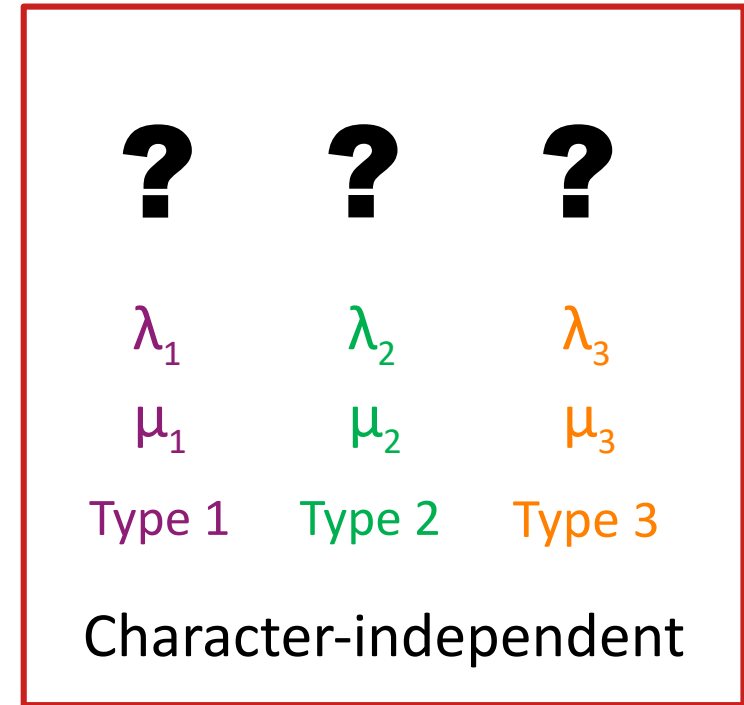


- Hybrid model with a hidden trait (A/B) added to the user-defined trait (1/2)
- Allows to distinguish whether the user-chosen character is linked to the rate variation
- Remaining issues:
 - The number of values for the hidden trait is chosen by the user
 - Higher complexity of the model

Character-dependent or independent ?



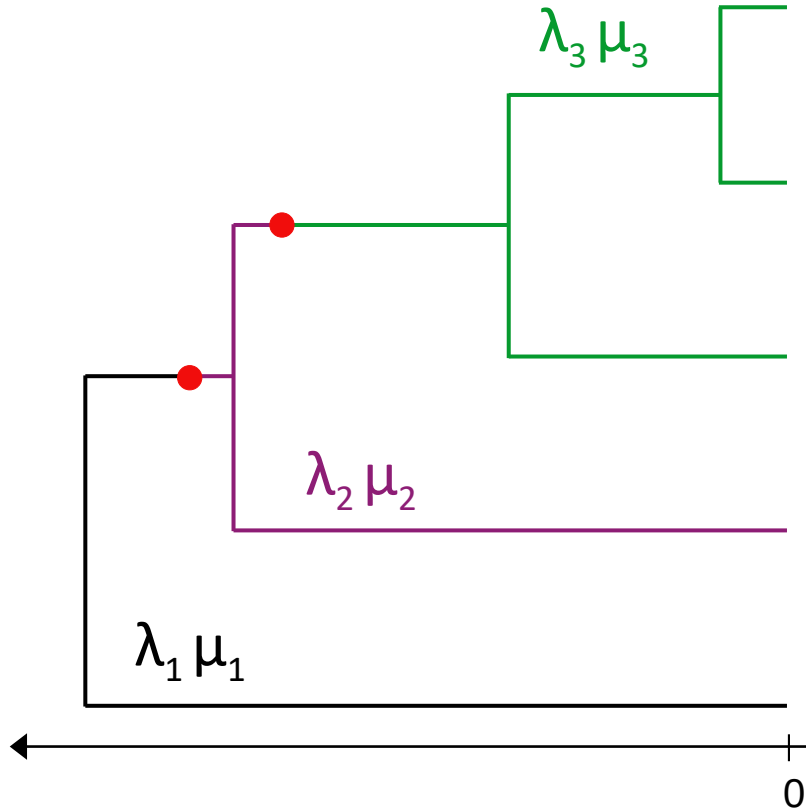
Character-dependent



In a character-dependent model :

- The number of types is known
- The type at the tips is known

BAMM/MSBD model

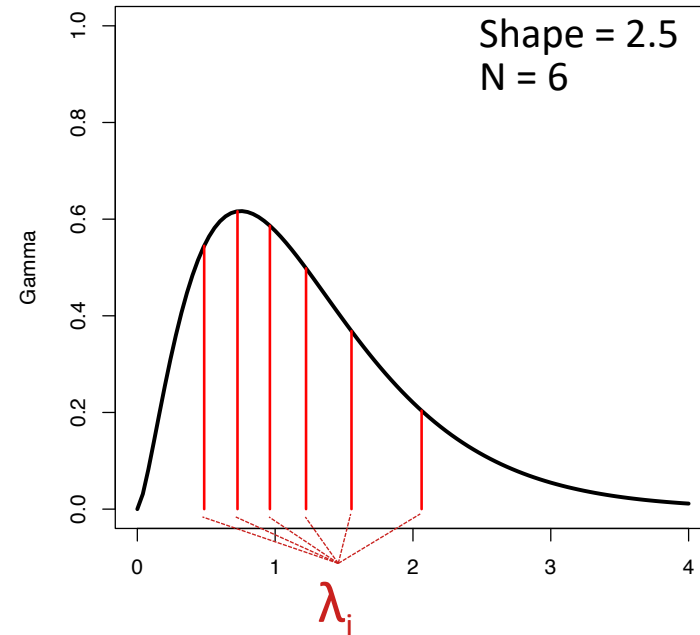
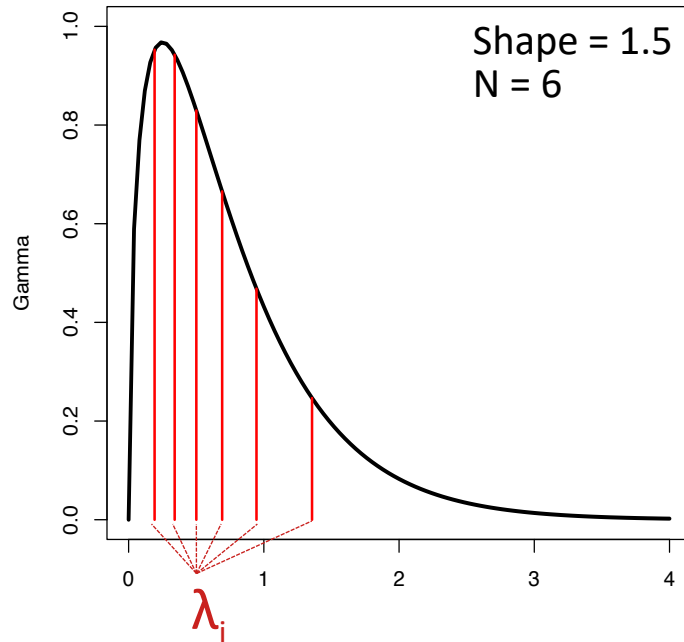


- Character-independent version of SSE
- New estimated parameters:
 - N total number of types
 - Types of edges and tips
- Simplified transition process: ●
 - Each transition is a new type (BAMM)
 - Constant transition rate γ (MSBD)

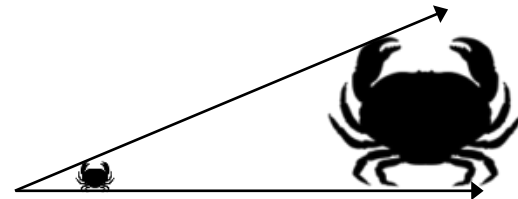
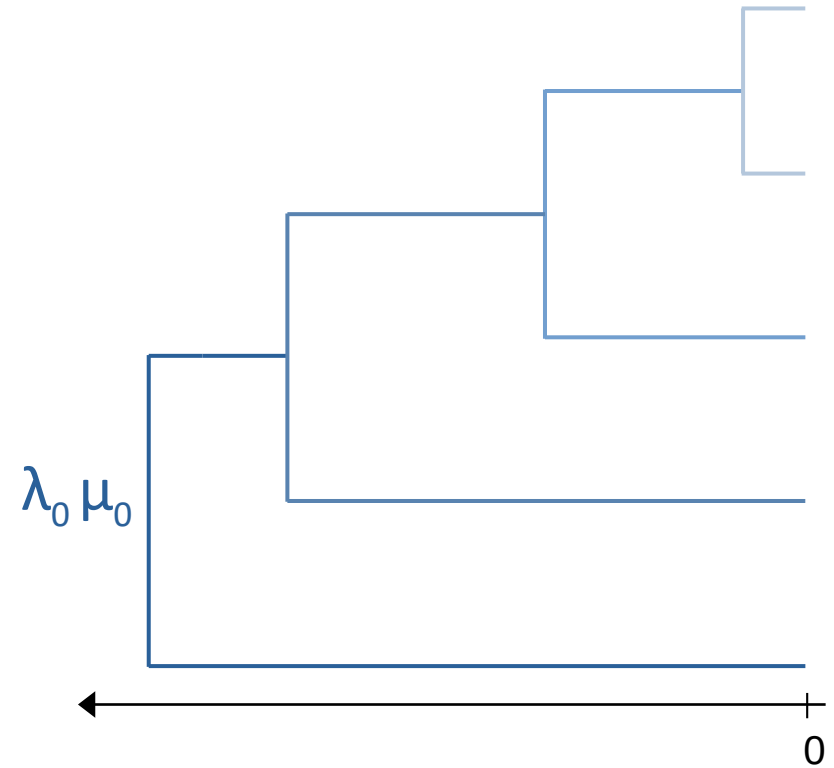
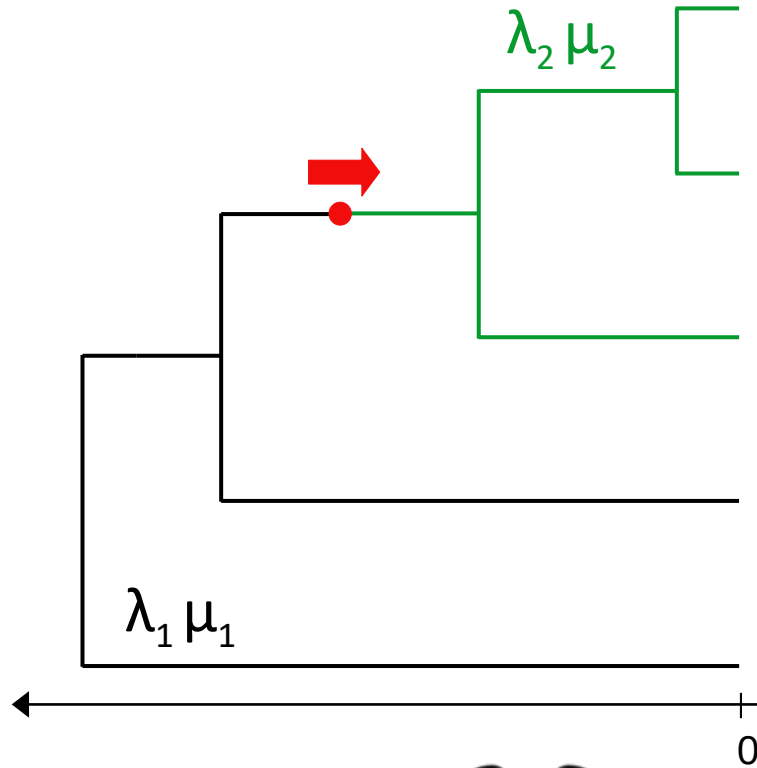
Rabosky *et al.* **Nat. Comm.** 2013
Barido-Sottani *et al.* **Sys. Bio.** 2020

RevBayes model

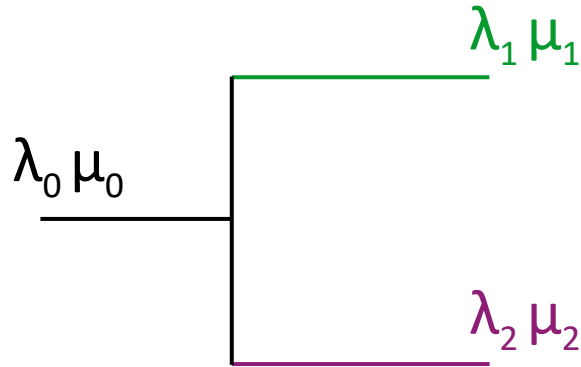
- Ordered types based on a Gamma distribution
- Fixed number of types N
- Simplified model: rates are not estimated, but determined by the shape of the Gamma distribution



Going beyond types



ClaDS model



$$\lambda_1 = \text{LogNormal}(\lambda_0 \times \alpha, \sigma)$$

$$\lambda_2 = \text{LogNormal}(\lambda_0 \times \alpha, \sigma)$$

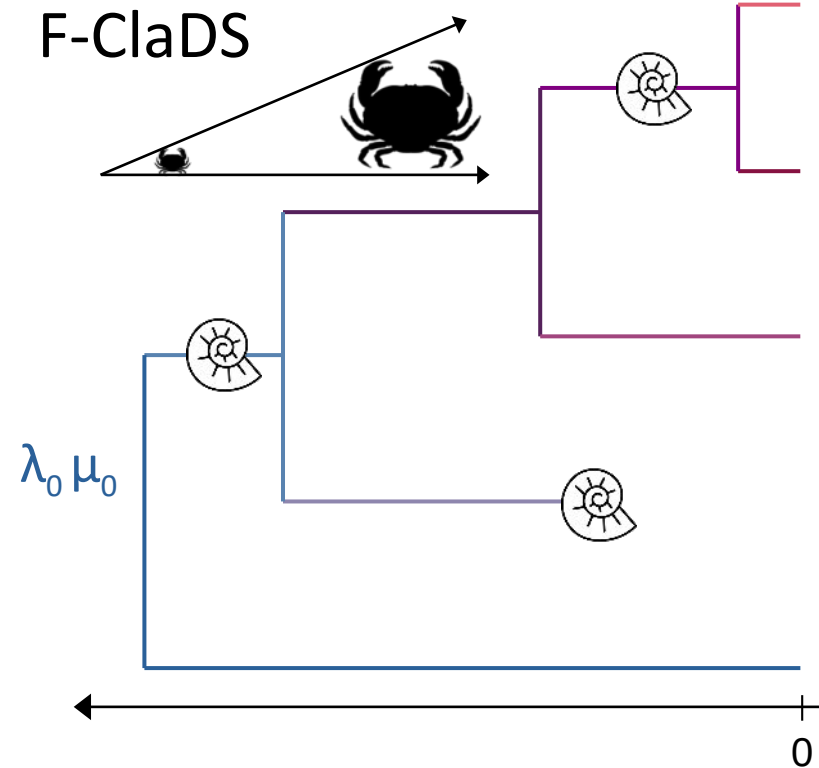
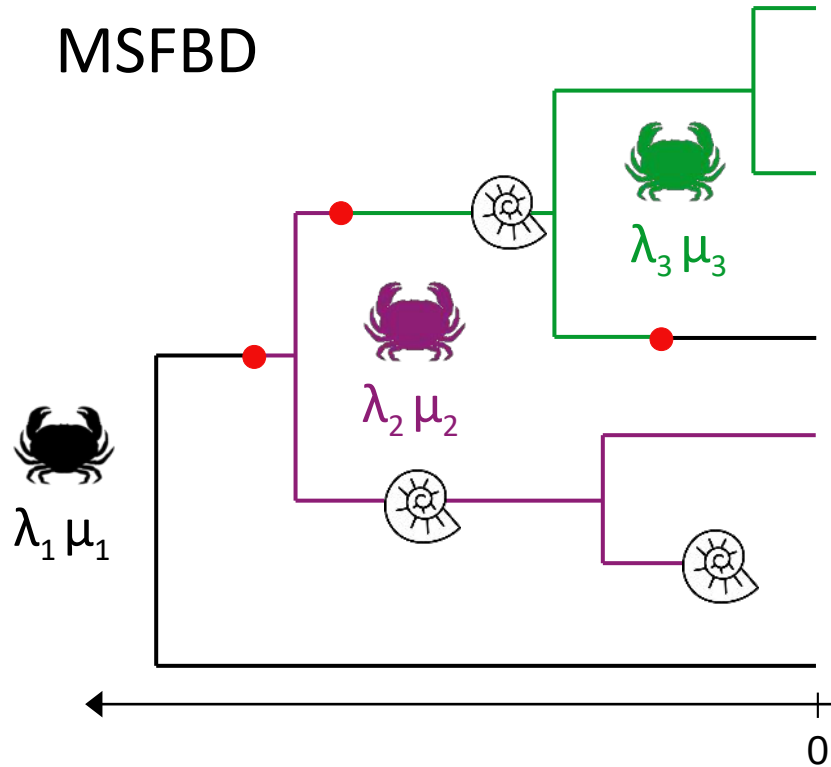
- Continuous evolution process driven by a lognormal distribution
- Inheritance / autocorrelation process: descendant rates partially inherited from the ancestral rates
- New estimated parameters:
 - Initial rates at the root λ_0 and μ_0
 - Lognormal parameters α and σ
 - Birth rates for each edge λ_i

Maliet *et al.* **Nat. Eco. Evo.** 2019

Maliet & Morlon **Sys. Bio.** 2021

Barido-Sottani & Morlon **Sys. Bio.** 2023

Heterogeneous FBD models



Summary

- FBD phylogenetic inference is a **powerful** and **flexible** tool for integrating fossil samples/species into phylogenies
- FBD phylogenetic inference is an **active** area of development
 - More complete integration of existing data
 - More realistic models of evolutionary processes
- Important challenges remain
 - Computational performance
 - Identifiability of models faced with limited amounts of data
 - Accurate modelling of morphological information